SellerHealth Watchlist

A Lightweight, Explainable Seller Risk Monitoring System for Marketplace Seller-Services Teams

GitHub: github.com/thedikshantmahlawat/sellerhealth-watchlist

Problem Statement

On any seller marketplace, a small share of sellers drives a disproportionate share of late shipments, cancellations, and refund/customer-service cost. Without a systematic way to score and rank sellers on this, that concentration stays invisible until it surfaces as complaint volume, policy strikes, or buyer churn. This project builds a seller-level early-warning system: a composite risk score that an operations manager can audit line by line, and a separate anomaly flag for sellers who break down suddenly rather than gradually. Both are deliberately simple rather than a black-box model — a seller coaching program only works if the people running it trust and understand the score behind it.

Approach & Methodology

The pipeline runs in four stages: a documented SQL aggregation layer builds seller-month metrics (late shipment rate, cancellation rate, review score, revenue) from order-level data; a Python layer computes a transparent weighted risk score plus a rolling-anomaly flag; a Streamlit dashboard makes both explorable; and a Power BI export layer turns the same output into a clean star schema for BI-tool reporting.

Risk score weighting (0–100 scale, higher = higher risk):

Signal	Weight	Why this weight
Late shipment rate	35%	Largest driver of CS tickets & refunds; fully within seller control
Cancellation rate	30%	Direct lost revenue + refund cost; usually a stock-accuracy issue
Review score (inverted)	25%	Real but lagging signal, blended with product quality — weighted third
Order volume decline	10%	Early-warning sign, but swings also have benign causes (seasonality)

Each component is scored relative to that month's other sellers, not a fixed historical anchor — so a marketplace-wide disruption (e.g. a carrier delay) doesn't automatically flag every seller as high risk. The anomaly flag is a separate, second signal: it compares a seller's current score to their own recent trailing average using a pooled standard deviation (chosen after testing showed a naive per-seller rolling z-score over-flagged on small sample sizes), so a seller who was fine for a year and just broke down gets caught early, before it becomes chronic.

Key Findings (demo data)

Figures below are computed by the pipeline on the project's data (see “A note on the data”) over the trailing 3-month window in that dataset. Currency in BRL (R$), the dataset's native currency.

Metric (trailing 3-month window, 2018-04 to 2018-06)	Value
Sellers in bottom 5% by risk score	11 of 220
Cancellations attributable to that 5%	15.9%
Late shipments attributable to that 5%	11.2%
Sellers recommended for coaching (bottom 10%)	22
Seller-months flagged for sudden deterioration	28
Illustrative prevented refund/CS cost (window)	~R$ 1,291
Illustrative prevented cost, annualized	~R$ 5,163

The prevented-cost figures are an illustrative scenario calculation — they rest on one clearly labeled, adjustable assumption (an average cost per cancellation) rather than an audited real business number. The *calculation and that assumption are both in src/generate_business_case.py, ready to be swapped for a real internal cost figure.*

Tech Stack

- SQL (SQLite): documented aggregation from order-level to seller-month grain
- Python (pandas, numpy): weighted risk score, pooled-std rolling anomaly detection
- Streamlit: 3-module interactive dashboard (marketplace overview, ranked watchlist with trend sparklines, seller drill-down)
- Power BI: star-schema CSV export (fact + 3 dimension tables) with a full data dictionary and ready-to-use DAX measures

Relevance to Seller-Services Business Analyst Roles

This project was built to mirror the core loop of a marketplace seller-services BA role: turn raw order/seller data into a metric definition an ops team can act on, translate that into a ranked, explainable score, and package it for both a self-serve dashboard and a BI tool a stakeholder already uses. The explicit weighting rationale, the documented denominator choices in the SQL layer, and the clearly-labeled assumptions in the business-impact estimate are meant to reflect exactly the kind of scrutiny a real seller-performance metric gets before it drives a policy or a coaching program — the same scrutiny relevant to marketplace seller-operations analytics work of the kind done at companies like Amazon (seller services), Meesho, or in a risk/operations-analytics context at a financial services firm like JPMorgan Chase.